

C++ - Allocation mémoire et structure RAII

BTS CIEL



Rappel stack VS heap en C

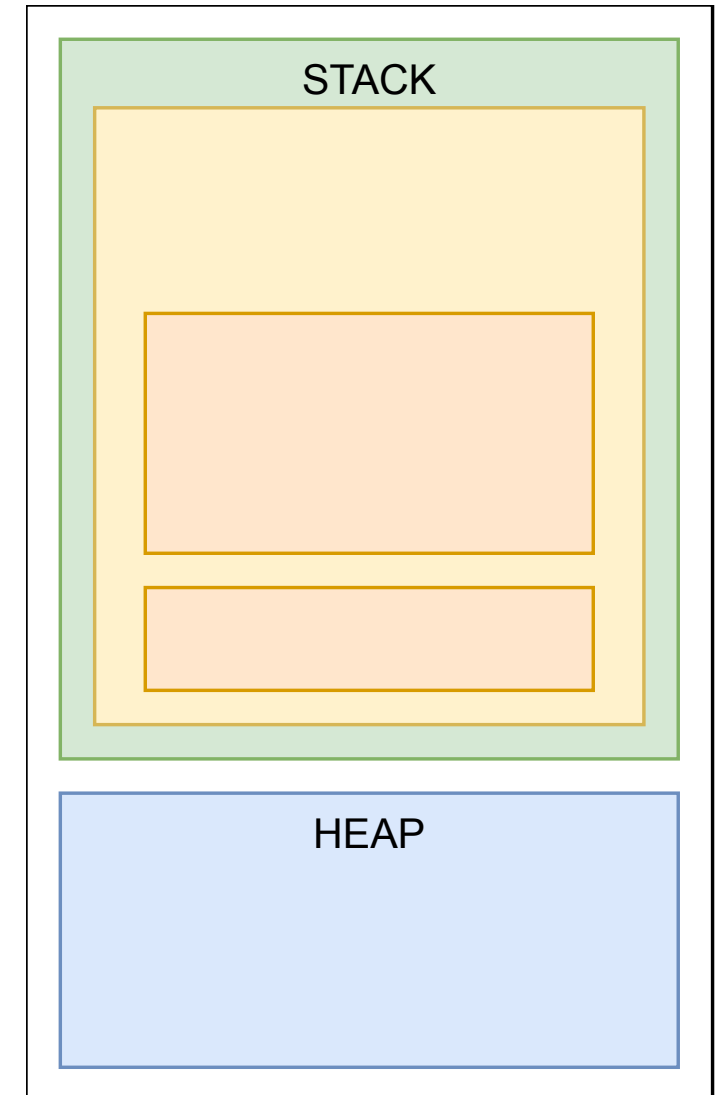
```
void add(int a, int b)
{
    return a + b;
}

int main()
{
    int a = 10;
    int b = 12;

    int result = add(a, b);

    printf("Result: %d", result);

    return 0;
}
```



Rappel stack VS heap en C

```
typedef struct Point {int x; int y;} Point;

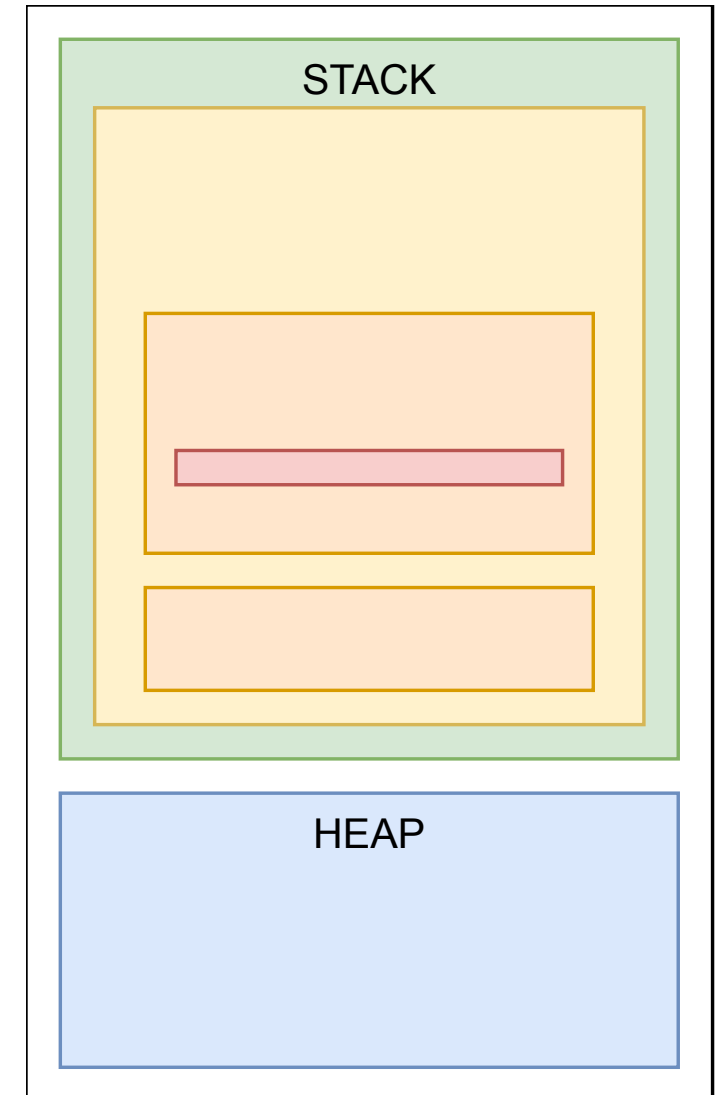
Point* new_point_from(Point a, Point b)
{
    Point* p = malloc(sizeof(Point));

    if (p == NULL) return NULL;

    p->x = a.x + b.x;
    p->y = a.y + b.y;

    return p;
}

int main()
{
    Point a = {10, 10};
    Point b = {20, 20};
    Point* p = new_point_from(a, b);
    free(p);
}
```



C++ new et delete

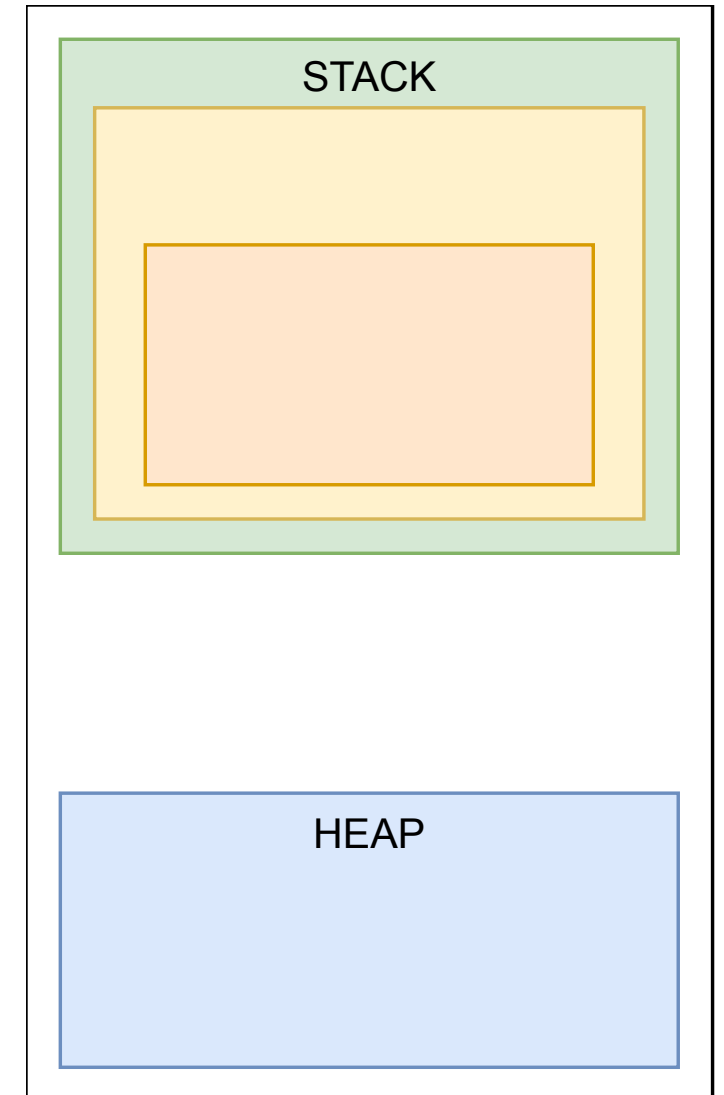
```
struct Point {int x; int y;};

Point* new_point_from(Point a, Point b)
{
    Point* p = new Point();

    p->x = a.x + b.x;
    p->y = a.y + b.y;

    return p;
}

int main()
{
    Point a = {10, 10};
    Point b = {20, 20};
    Point* p = new_point_from(a, b);
    delete p;
}
```



C++ référence

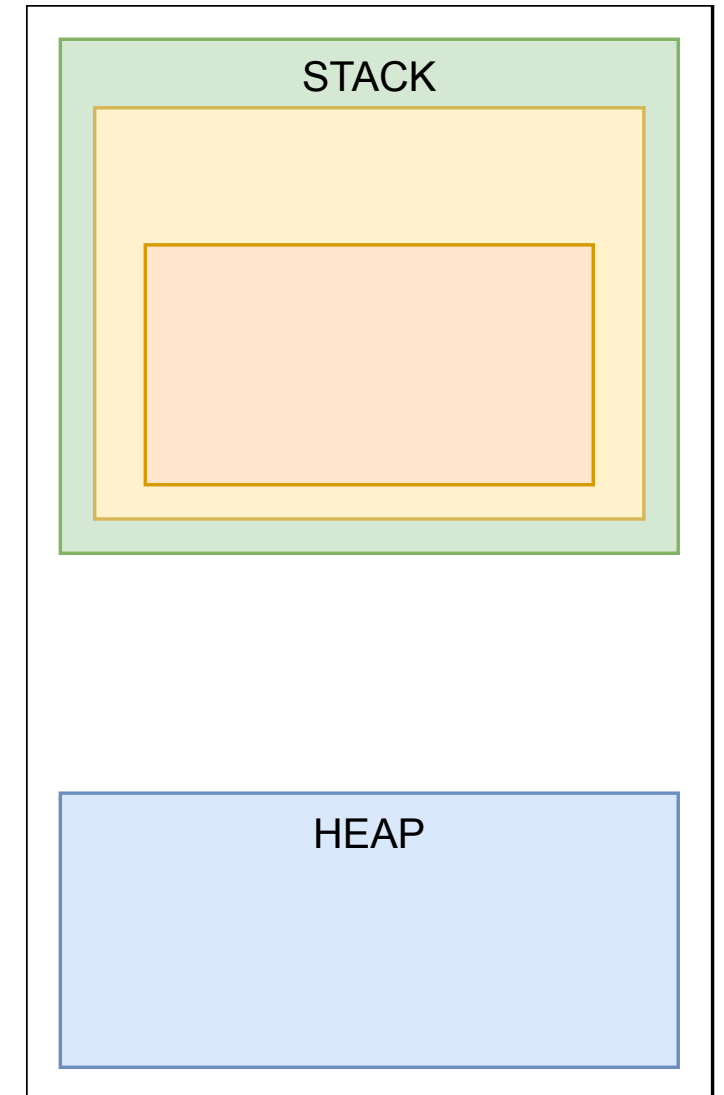
```
struct Point {int x; int y;};

Point* new_point_from(Point& a, Point& b)
{
    Point* p = new Point();

    p->x = a.x + b.x;
    p->y = a.y + b.y;

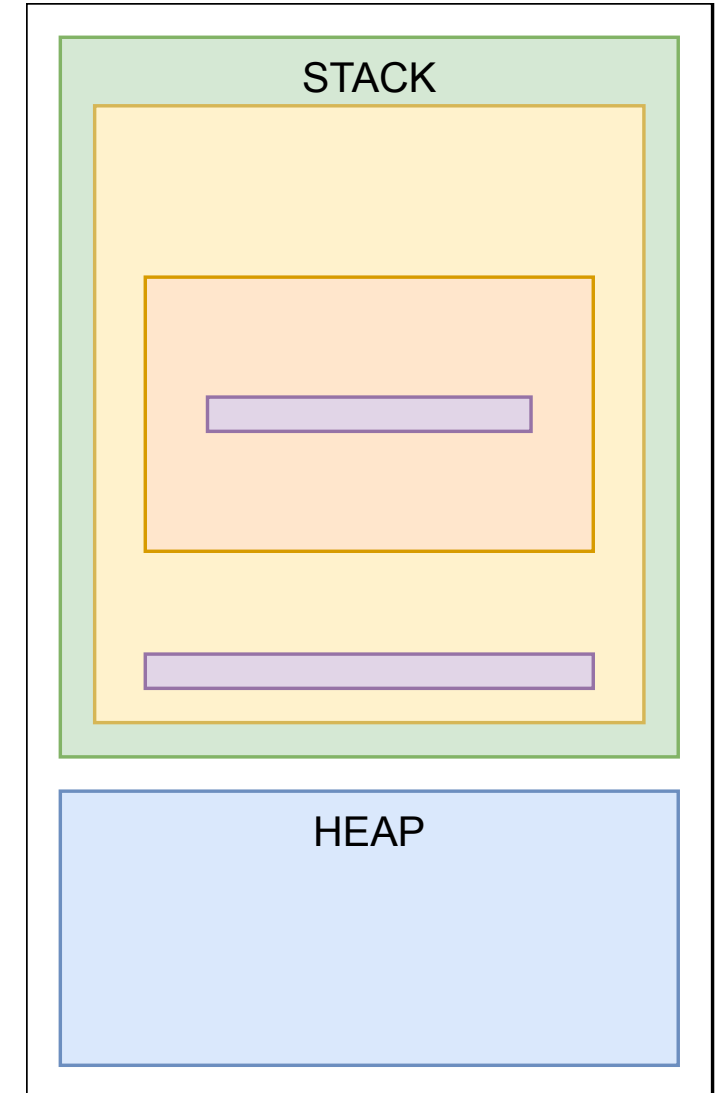
    return p;
}

int main()
{
    Point a = {10, 10};
    Point b = {20, 20};
    Point* p = new_point_from(a, b);
    delete p;
}
```



C++ constructeur et destructeur

```
struct Point {  
    int x; int y;  
  
    Point(int _x, int _y) : x(_x), y(_y) {}  
    ~Point() {}  
};  
  
Point* new_point_from(Point& a, Point& b)  
{  
    Point* p = new Point(0, 0);  
  
    p->x = a.x + b.x;  
    p->y = a.y + b.y;  
  
    return p;  
}  
  
int main()  
{  
    Point a = {10, 10};  
    Point b = {20, 20};  
    Point* p = new_point_from(a, b);  
    delete p;  
}
```



C++ RAII

```
struct Point {int x; int y;};

int main()
{
    std::vector<Point> points = {};

    points.push_back(Point(10, 5));
    points.push_back(Point(20, 13));
}
```

